

Employee Attrition Prediction - Executive Summary

To: HR Director

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Project Overview:

We analyzed our HR records (1,470 employees) to uncover why employees leave and to build a predictive system that flags 'at-risk' staff. Currently, our company loses about 16% of employees per year, with resignations peaking during the critical first year and fifth year of tenure.

Key Findings (What drives people to leave?):

- Overworked Staff: Employees working excessive overtime are at an incredibly high risk of burning out and leaving.
- Vulnerable Departments: The Sales department is our biggest pain point, specifically 'Sales Representatives' who have an alarming exit rate of nearly 40%.
- Salary is a factor, but not the whole story: While departing employees do earn less on average, factors like poor work-life balance and travel requirements strongly amplify their decision to quit.

Model Results:

We successfully built an early-warning system. Our best model reliably identifies the majority of employees who are likely to quit before they hand in their notice.

Strategic Recommendations:

1. Overtime Intervention: Implement a mandatory review for employees clocking excessive overtime.
Offering schedule flexibility or additional time-off can directly reduce turnover costs.
2. First-Year Mentorship: The first 12 months are highly volatile. Introduce a structured 'buddy' or mentorship program targeting new hires in Sales to anchor them to the company culture early.

Note on System Usage:

The system errs on the side of caution. It may occasionally flag loyal employees as 'at-risk'. Therefore, HR should use these flags as a trigger for a positive, supportive 'check-in' conversation rather than assuming the employee is already disengaged.